

ANUBHAB DAS

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EDUCATION

Guru Nanak Institute of Technology

B.Sc. in Cyber Security

Kolkata, India

Aug 2024 – May 2028

- Class Representative, B.Sc. Cyber Security Department (2024–Present)
- Participated in 3+ national hackathons including the **ISRO Hackathon** and SkillHigh Web Dev Hackathon

PUBLICATIONS

Blockchain-Based IoT Security: A Comprehensive Review of Architectures, Challenges and Future Directions *Dec 2025* *IEEE ICRCICN Conference (IEEE Indexed) — Co-author*

- Analyzed and synthesized blockchain-based security frameworks for IoT ecosystems; evaluated consensus mechanisms, trust architectures, and decentralized data integrity models across 50+ papers.

A Systematic Review of Blockchain-Based Security Architectures for Trusted Cloud Data Sharing *Apr 2026* *Journal of Techniques–Sciences–Methodes (TSM), Vol. 10, Issue 4 — Co-author*

- Evaluated hybrid blockchain architectures, smart contract access control, decentralized identity, and privacy-preserving encryption against threat models including Sybil and insider attacks.

Hardware-Aware Co-Design Framework for Efficient TinyML Deployment on Ultra-Low-Power Edge Devices *Jun 2026* *Book Chapter — Co-author*

- Investigated hardware-aware model compression, quantization, structured pruning, and adaptive runtime scheduling for TinyML on microcontroller-class devices.

Offline-First TinyML–Federated Learning Framework for Privacy-Preserving Smart Healthcare Systems *Jun 2026* *Book Chapter — Co-author*

- Proposed a federated learning architecture integrating on-device TinyML inference with differential privacy for energy-efficient, offline-capable smart healthcare edge systems.

RESEARCH INTERESTS

Cybersecurity • Blockchain & Distributed Systems Security • IoT Security • AI/ML Security

EXPERIENCE

Infotact Solutions

Associate L1 — Web Development Intern

Remote

Aug 2025 – Oct 2025

- Conducted secure code reviews against **OWASP Top 10**; identified and remediated injection, XSS, and CSRF vulnerabilities across client web applications.
- Developed and tested modular front-end components; improved load performance and defect detection rate through structured cross-browser QA.

Prodigy InfoTech

Full-Stack Developer Intern

Remote

Jul 2025

- Built **Flask** REST API endpoints with server-side input validation, parameterized queries (SQLi prevention), and token-based authentication; received **Letter of Recommendation** for high-performance delivery.

Internship Studio

Website Design & Development Intern

Remote

Aug 2024 – Sep 2024

- Delivered WCAG-compliant, responsive UIs with client-side validation; implemented UX and SEO enhancements improving mobile and desktop usability.

PROJECTS

NexusScan | Python, Flask, JS, REST APIs, ML, LLM Integration | [NexusScan](#)

2025

- Built a real-time URL threat-intelligence platform with multi-engine risk scoring, ML-based phishing detection, QR scanning, and LLM-generated threat explanations (Groq / Gemini / OpenAI).
- Hardened backend security: fixed encoding bugs, recalibrated risk scoring, enforced CORS policy, and sanitized IP extraction; deployed on Vercel with `.env`-only configuration.

CyberASCII | JavaScript, Node.js, WebRTC, Gemini AI API, Canvas API

2026

- Developed an AI-powered cybersecurity visualization tool converting live camera feeds and images to real-time ASCII analysis using on-device processing and privacy-first architecture.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML5, CSS3, SQL, Bash

Frameworks & Libraries: Flask, Bootstrap, Chart.js, React (learning), Node.js (learning)

Security: OWASP Top 10, Threat Modeling, Threat Intelligence, Application Security, Secure SDLC, XSS/CSRF/SQLi Prevention, Secure Code Review, Input Validation

Research Areas: Systematic Literature Review, Blockchain Security, IoT Security, Privacy-Preserving ML, Federated Learning, TinyML, Edge Intelligence

Tools & Platforms: Git, GitHub, Linux, VS Code, Vercel, REST APIs, Browser DevTools

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Mastercard Cybersecurity Job Simulation • Tata Cybersecurity Analyst Job Simulation • Cisco Certificate • JavaScript Bootcamp • Introduction to CIP

Achievements: 4 research publications (IEEE, peer-reviewed journal, book chapters) within Year 2 of undergraduate study • ISRO Hackathon participant • Class Representative, Cyber Security Dept.